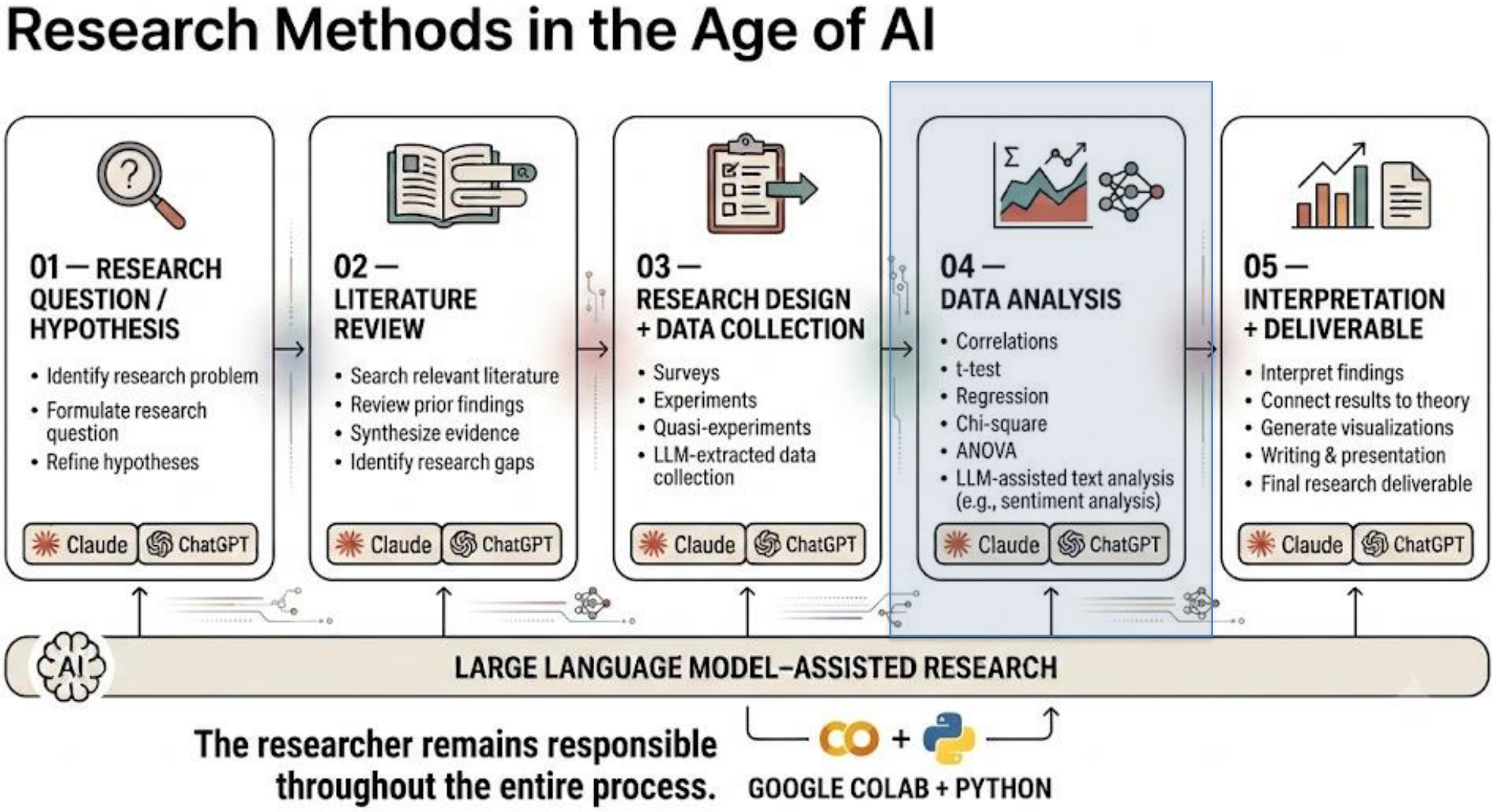
BLOCK 3

From Research Design to Data Analysis: Data Cleaning and Preparation

Course Roadmap and Syllabus



Block 2 recap: the decisions you make BEFORE analysis

Survey	Measure variables as they naturally occur; useful for description and association.
Experiment / A/B test/Randomized controlled trials	Manipulate an IV and randomly assign conditions; strongest for causal inference.
Quasi-experiment	Compare an intervention/exposure without full random assignment; alternative explanations matter more.

MEASUREMENT
How do we turn a construct into a variable?

RELIABILITY
Would the measure be reasonably consistent?

VALIDITY
Does the evidence support the interpretation we want to make?

Why are we learning this?

- Block 2 taught you how to design and collect evidence.
- Block 3 asks: once the responses are in a dataset, how do we turn them into evidence?
- A statistical test cannot rescue a poor design — and a good design can still be undermined by data problems.

Today is the bridge between DESIGN and STATISTICS.

Our Study Design: From Participants to Data

A randomized experiment with two conditions and a post-message survey (N = 180).

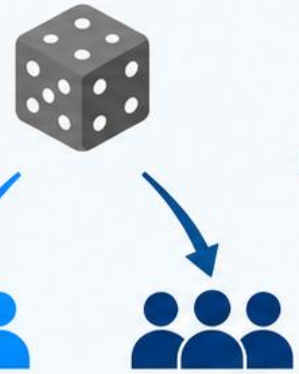
1 Participants

180 people (e.g., USC students) are recruited to participate in an online study.



2 Random Assignment

Each participant is randomly assigned to one of two conditions (like flipping a coin).



Control
(standard message)
~ 90 participants

Personalized
(adapted message)
~ 90 participants

3 View the Message

Participants see a social media post based on their assigned condition.

Control condition (standard message)



Personalized condition (adapted to the user)



4 Complete the Survey

After viewing the message, participants answer a short survey.

Message trust
(1 = strongly disagree, 7 = strongly agree)

○ 1 ○ 2 ○ 3 ○ 4 ○ 5 ○ 6 ○ 7

Intention to share
(1 = strongly disagree, 7 = strongly agree)

○ 1 ○ 2 ○ 3 ○ 4 ○ 5 ○ 6 ○ 7

Clicked the article?

○ Yes ○ No

Demographics (examples)

- Age
- Gender
- Education level
- Primary social media platform
- Daily social media use (hours)

5 Resulting Data

Each row in the dataset represents one participant's responses and characteristics (N = 180 rows).

ID	Condition	Age	Trust	Share	Clicked	...
1	Control	22	4.2	3	No	...
2	Personalized	24	5.8	5	Yes	...
3	Control	21	3.9	2	No	...
4	Personalized	27	6.1	6	Yes	...
5	Control	19	2.7	1	No	...
...	...	...	...	...	...	...

What is a dataset?

The new course dataset includes experimental, demographic, media-use, and outcome variables.

ROW = one participant / observation

COLUMN = one variable

ID	condition	age	gender	platform	message_trust	clicked
1	Control	24	Woman	TikTok	4.2	No
2	Personalized	31	Man	YouTube	5.8	Yes
3	Control	27	Woman	Instagram	3.9	No

A column can be demographic, experimental, behavioral, or an outcome — its ROLE depends on the research question.

Recap: Reading Python Code

Symbol	Think of it as…	Simple example
`()`	**Do something / give something information**	`print("Hello")`
`[]`	**A group of things OR pick something from a group**	`names[0]`
`.`	**Use something that belongs to something else**	`text.upper()`
`" "` or `' '`	**This is text**	`"Hello"`
`,`	**Separate things**	`print("Kay", 10)`
`#`	**A note for humans — Python ignores it**	`# My analysis`
`=`	**Give something a value**	`age = 25`
`==`	**Ask: Are these equal?**	`age == 25`

Quick Python recap: how to read the code

You are not memorizing syntax. Read code from left to right and ask what object, column, and action you see.

```
df["message_trust"].mean()
```

df → choose column → calculate mean

```
df.groupby("condition")["message_trust"].mean()
```

group by condition → select trust → mean

- Parentheses ( ) usually mean “do this action.”
- Square brackets [ ] usually select a column or subset.
- Quotation marks identify a column name exactly as it appears in the dataset.

IV, DV, and Predictors: What's the Difference?

These terms describe the role a variable plays in your research question and analysis.

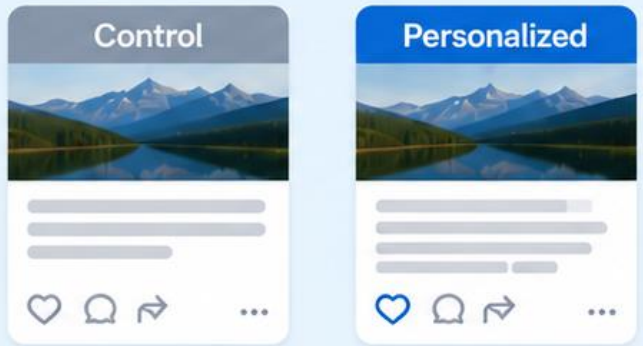
Independent Variable (IV)

The factor you change or compare.

- The cause or treatment in an experiment (or the group you compare).
- You decide the values (e.g., control vs. personalized).
- Often used to see if it leads to differences in an outcome.

Example from our study

Condition (message type)
(Control vs. Personalized)



Dependent Variable (DV)

The outcome you measure.

- The result or outcome you want to explain or predict.
- Measured after the IV (e.g., after participants view the message).
- In our study, these are the survey responses.

Examples from our study (dependent variables):

message_trust
How much participants trust the message.
(1 = strongly disagree, 7 = strongly agree)

share_intention
Intention to share the message.
(1 = strongly disagree, 7 = strongly agree)

clicked_article
Whether the participant clicked the linked article.
(Yes / No)

Predictors

Variables used to explain or predict an outcome.

- Any variable that may help explain differences in the DV.
- Includes the experimental condition and other characteristics (e.g., demographics, behavior, attitudes).
- Used in analyses like regression to see which variables are related to or predict the outcome.

Examples from our study (predictors)

condition	Experimental condition (Control vs. Personalized)
age	Participant's age (in years)
gender	Gender identity
education_level	Highest education level
primary_platform	Main social media platform
daily_social_media_hours	Hours of social media use per day
baseline_trust	Pre-existing trust in news/media (1 = strongly disagree, 7 = strongly agree)

Variable type and variable role are different questions

NUMERIC

age = 31
message_trust = 5.8
daily_social_media_hours = 3.2

Arithmetic makes sense.

CATEGORICAL

gender = Woman
condition = Control
platform = TikTok
clicked_article = Yes

Values represent groups.

IDENTIFIER

participant_id = 37

37 is a label — participant 37 is not “more” than participant 36.

Usually not IV or DV.

Before choosing a statistical test: What TYPE is each variable, and what ROLE does it play in my question?

Demographics can be predictors — but “IV” and “DV” come from the question

Example A
Does message trust differ by education level?

IV / grouping variable = education_level
DV / outcome = message_trust

Example B
Is age associated with message trust?

Predictor = age
Outcome = message_trust

Example C
Does personalization work differently for younger vs. older participants?

IV = condition; moderator/predictor = age
DV = message_trust

Example D
What predicts daily social-media use?

Possible predictors = age, gender, employment
DV / outcome = daily_social_media_hours

A demographic variable is not automatically an IV. The research question determines its analytical role.

Messy data are normal

Cleaning is not “making the data say what you want.” It is identifying and documenting problems.

MISSING

participant skipped a question
NaN / blank

INCONSISTENT

“Instagram” vs “instagram”
same meaning, different label

IMPOSSIBLE

99 hours/day
9 on a 1–7 scale

DUPLICATE

same participant appears twice
may double-count a case

Important: every cleaning decision should have a reason.

Step 1 — Load the SAME course dataset

```
import pandas as pd

url = "https://raw.githubusercontent.com/KayChansiri/Course_research-methods-in-the-age-of-ai/main/data/JOUR701_Block3_Course_Dataset.csv"

df = pd.read_csv(url)

df.head()
```

- pandas = a library for working with tables
- df = the name of our DataFrame
- read_csv = read a CSV file from a file path OR a web URL

Step 2 — Look before you analyze

```
df.head()

df.shape

df.columns

df.info()
```

- What does one row represent?
- What does each column represent?
- How many rows and columns are there?
- Did Python read each variable as the type you expected?

Never start with a test just because you know the code.

What Python says	Simple meaning	Example from our dataset	Usually corresponds to
int64	Whole numbers	participant_id	Numeric or identifier
float64	Numbers that can have decimals	age, message_trust	Usually numeric
object	Text / labels	condition, gender	Usually categorical

Step 3 — Check missing data

```
df.isna().sum()
```

- Missing does not automatically mean “bad data.”
- Possible decisions: keep the case, remove a row for a specific analysis, investigate why it is missing, or use a more advanced method.
- For today only, we will remove rows missing our main outcome.

```
df = df.dropna(subset=["message_trust"])
```

This changes the DATAFRAME — not the original CSV.

Step 4 — Impossible values are not the same as outliers

UNUSUAL BUT POSSIBLE

8 hours/day of social media

Could be real. Investigate before deleting.

IMPOSSIBLE / INVALID

99 hours/day OR 9 on a 1–7 scale

Violates what the variable can represent.

```
df["daily_social_media_hours"].describe()
```

Python will calculate with 99 unless YOU notice it.

Step 5 — Check BOTH variable types and values

A variable can have the wrong label, the wrong value, or the wrong Python data type.

```
df.dtypes
```

Ask: Should this column be numeric, categorical/text, or an ID?

```
df["condition"].value_counts()
df["primary_platform"].value_counts()
```

Find inconsistent labels:
Personalized vs personalized
Instagram vs instagram

```
df["condition"] = df["condition"].str.title()
df["age"] = pd.to_numeric(df["age"], errors="coerce")
```

Step 6 — Does the value fit the measurement?

The survey says share_intention must be between 1 and 7.

First, look at the range

```
df["share_intention"].min()
df["share_intention"].max()
```

Maximum = 9?!

Then, show the invalid row

```
df[df["share_intention"] > 7]
```

Why this is easier: first ask “What is the minimum/maximum?” Then inspect only the values that violate the known 1–7 scale.

Step 7 — Describe the cleaned data

```
df.describe()

df["condition"].value_counts()

df.groupby("condition")["message_trust"].mean()
```

- Descriptive statistics tell us what is in our sample.
- Group means can show a pattern, but they do not by themselves tell us whether the difference is statistically convincing.
- Later, the t-test will address that inferential question.

What is an Assumption?

An **assumption** is something that needs to be reasonably true about your data for a statistical analysis to work properly.

1 Simple idea

An assumption is a **check before you analyze** to make sure your data make sense and fit what the analysis expects.

2 Examples in our dataset

Values are reasonable
(no obvious errors)

Example (age):
22, 24, 31 ... ✓
Not 222 or −5 ✗

Categories are coded consistently

Example (condition):
"Control" and "Personalized" ✓
Not "control", "CONTROL" ✗

Missing values are checked

Example (age):
178 out of 181 have values ✓
3 are missing (NaN)

Data are in the right format

Example (clicked_article):
"Yes" or "No" ✓
Not mixed text like "Y", "Yes", "yes", "1" ✗

3 Why it matters

If the data do not meet the needed assumptions, the results can be incorrect or misleading.

Checking assumptions helps you choose the right analysis and trust your results.

It makes your research more accurate and credible.

Key takeaway

Before we run a statistical test, we **check whether our data are appropriate for that test.**

You do not need to memorize specific assumptions now. We will learn the important assumptions when we learn each statistical method.

Four assumption ideas — visually

INDEPENDENCE



separate observations

Ask: Is one participant's response being counted as if it were another independent person?

VARIABLE TYPE

123 vs A/B

Numeric outcome?
Categorical groups?

The test must match what the variables represent.

DISTRIBUTION



Look for extreme skew or strange values in a numeric outcome.

VARIANCE

Group A: ••••
Group B: • •

Some comparisons care whether group spreads are reasonably similar.

Later methods revisit the assumptions that matter for that specific test.

A visual check can reveal problems quickly

```
df["message_trust"].hist()
```

Histogram → shape, skew, unusual values

```
df.boxplot(column="message_trust",
           by="condition")
```

Boxplot → spread + possible outliers

- Look for strange values, extreme skew, very different group spreads, or patterns that deserve investigation.
- A plot does not “prove” an assumption. It helps you inspect the data.

When Python gives you an error: give ChatGPT enough information

1. GOAL
What were you trying to do?

2. CODE
Paste the exact code you ran.

3. ERROR
Paste the complete error message.

4. VERIFY
Run the fix and check whether it makes sense.

```
I am a beginner using pandas in Google Colab.
My goal is: [goal]
My code is: [paste exact code]
My complete error is: [paste error]
Explain it simply and give the smallest correction. Do not invent columns.
```

Debugging with ChatGPT: a concrete example

YOUR CODE

```
df["condition"].value_count()
```

PYTHON ERROR

AttributeError:
'Series' object has no attribute 'value_count'

LIKELY FIX

```
df["condition"].value_counts()
```

Ask ChatGPT: “Explain why this error happened, show the corrected line, and tell me what the corrected code does. Do not change anything else.”

Bad AI question vs. useful AI question

✗ TOO VAGUE

"My Python doesn't work. Fix it."

Missing:
- goal
- exact code
- complete error

AI may guess and rewrite too much.

✓ DEBUGGABLE

"I am using pandas in Colab. I want to count categories. Here is my exact code and complete error. Explain the error simply and give the smallest correction."

Then run it yourself and check the result.

AI can help debug code. It does not remove your responsibility to understand and verify the analysis.

In-class checkpoint

Work with the person next to you for 4 minutes.

- Find ONE problem in the course dataset.
- Show the code that helped you find it.
- Explain why it matters for the research — not just for Python.
- Be ready to tell the class what you found.

Connect the DATA problem back to DESIGN or MEASUREMENT.

What you need to leave with

- ✓ Rows are cases; columns are variables.
- ✓ Variable TYPE and analytical ROLE are different.
- ✓ Always inspect data before testing a hypothesis.
- ✓ Cleaning decisions require methodological reasons.
- ✓ Assumptions matter because statistical methods are models, not magic.
- ✓ For AI debugging: goal + code + full error + verify the fix.

Next class: you will do this yourself in Colab using the SAME course dataset.

What comes next? Same dataset, different questions

Control vs Personalized → message_trust	t-test
baseline_trust + message_trust	correlation
age + baseline_trust → message_trust	regression
4 platforms → message_trust	ANOVA
condition + clicked_article	chi-square